

Experiment No. 10

Title

Shell Programming Using Bash

Aim

The problem is to write and execute Bash shell scripts that demonstrate interactive input, positional parameters, arithmetic operations, conditional statements, and loop constructs, thereby automating basic computational and logical tasks.

Theory

Shell Programming

Shell scripting is a method of automating tasks by writing a sequence of Linux commands in a script file executed by the shell.

Arithmetic Operations

Arithmetic can be performed using:

`$((expression))`

Interactive Scripts

Interactive scripts accept input from the user during execution using the read command.

Positional Parameters

Shell scripts accept command-line arguments using:

- \$0 – Script name
- \$1, \$2, ... – Positional parameters

Conditional Statements

Decision-making constructs include:

- if
- if-else
- elif

Loops

Loops execute commands repeatedly:

- for
- while

COMMANDS USED-

read	Accept user input
echo	Display output
\$(())	Arithmetic operations
if, elif, else	Conditional statements
for, while	Loop constructs
chmod	Make script executable

Procedure / Steps

A. Interactive Shell Script

1. Create a script file: nano interactive.sh

2. Enter the following code:

```
#!/bin/bash  
  
echo "Enter your name:"  
  
read name  
  
echo "Welcome $name"
```

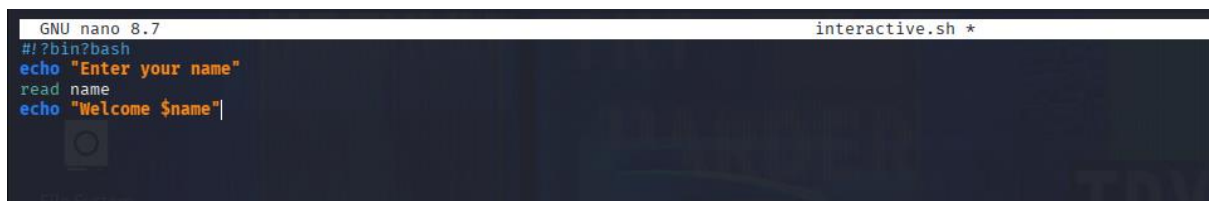
3. Save and exit.

4. Make the script executable:

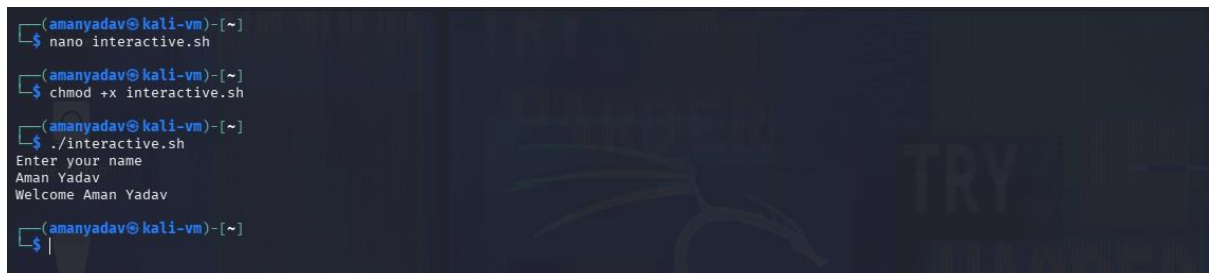
```
chmod +x interactive.sh
```

5. Execute the script:

```
./interactive.sh
```

A screenshot of the GNU nano 8.7 text editor. The title bar shows 'interactive.sh *'. The editor contains the following code:

```
#!/bin/bash  
echo "Enter your name"  
read name  
echo "Welcome $name"
```

A screenshot of a terminal window with the prompt '(amanyadav@kali-vm)-[~]'. The user enters 'nano interactive.sh', then 'chmod +x interactive.sh', and finally './interactive.sh'. The script's output is shown:

```
Enter your name  
Aman Yadav  
Welcome Aman Yadav
```

B. Using Positional Parameters

Create another script:

nano params.sh

Enter the following code:

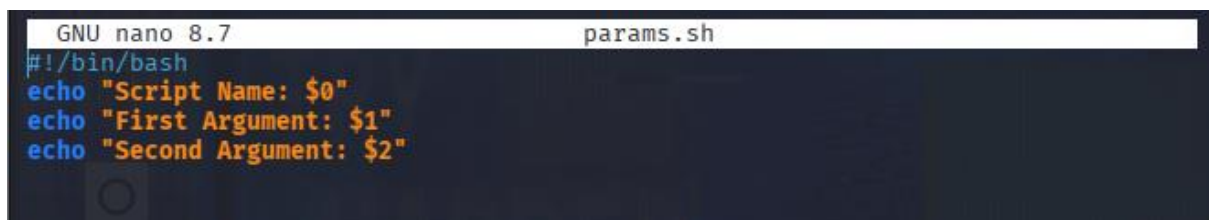
```
#!/bin/bash
```

```
echo "Script Name: $0"
```

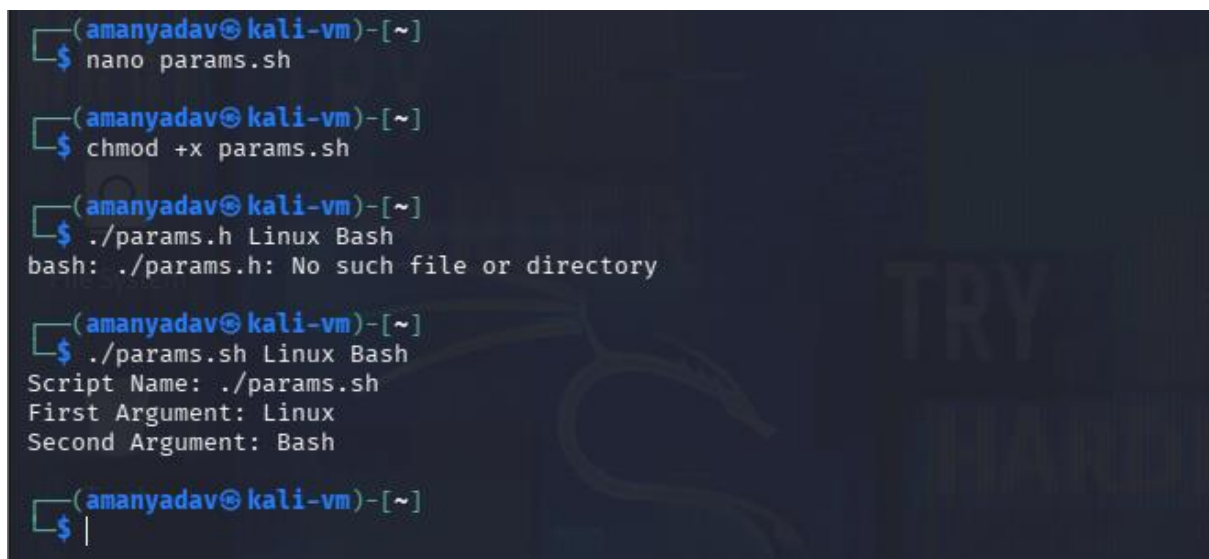
```
echo "First Argument: $1"
```

```
echo "Second Argument: $2"
```

Execute: ./params.sh Linux Bash

A screenshot of the GNU nano 8.7 text editor. The title bar shows 'GNU nano 8.7' and 'params.sh'. The editor contains the following text:

```
#!/bin/bash
echo "Script Name: $0"
echo "First Argument: $1"
echo "Second Argument: $2"
```

A screenshot of a terminal window with the prompt '(amanyadav@kali-vm)-[~]'. The user enters 'nano params.sh', then 'chmod +x params.sh'. They then enter './params.h Linux Bash', which results in an error: 'bash: ./params.h: No such file or directory'. Finally, they enter './params.sh Linux Bash', which outputs:

```
Script Name: ./params.sh
First Argument: Linux
Second Argument: Bash
```

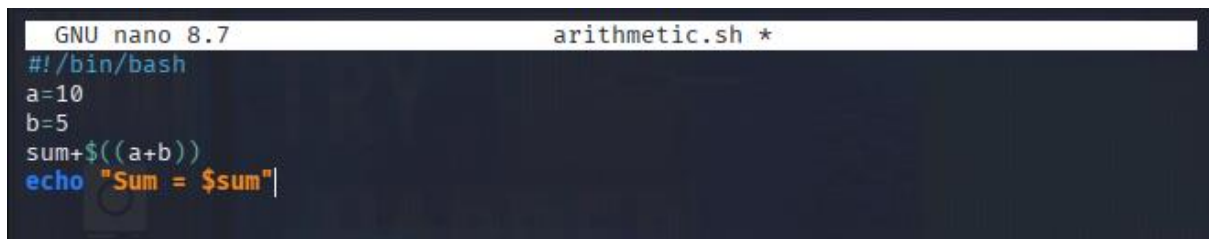
C. Arithmetic Operations

Create a new script file:

nano arithmetic.sh

Enter the following script:

```
#!/bin/bash  
  
a=10  
  
b=5  
  
sum=$((a + b))  
  
echo "Sum = $sum"
```



```
GNU nano 8.7 arithmetic.sh *  
#!/bin/bash  
a=10  
b=5  
sum=$((a+b))  
echo "Sum = $sum"
```

D. Conditional Statement

Create script:

nano condition.sh

Enter: #!/bin/bash

echo "Enter a number:"

read num

if [\$num -gt 0]

then

echo "Positive number"

else

echo "Zero or Negative number"

fi

```
GNU nano 8.7 condition.sh *
#!/bin/bash
echo "Enter a number:"
read num
if [ $num -gt 0 ]
then
    echo "Positive number"
else
    echo "Zero or Negative number"
fi
```

E. Loop Constructs

Create for-loop script:

nano loop.sh

Enter:

```
#!/bin/bash
for i in 1 2 3 4 5
do
    echo "Iteration $i"
done
```

```
GNU nano 8.7 loop.sh *
#!/bin?bash
for i in 1 2 3 4 5
do
    echo "Iteration $i"
done
```

Outcomes

- ✓ Scripts accept user input and display results correctly.
- ✓ Positional parameters are accessed successfully.
- ✓ Arithmetic operations produce correct output.
- ✓ Conditional statements and loops execute as expected.

Result

- ❖ Bash shell scripts demonstrating interactive input, positional parameters, arithmetic operations, conditional statements, and loops were successfully created and executed.